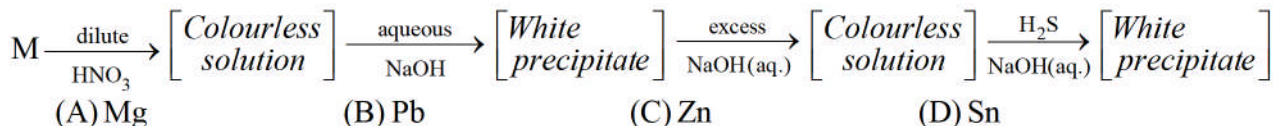


WORK SHEET - 04

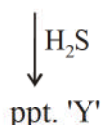
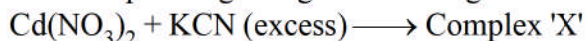
SALT ANALYSIS

Single correct :

Q.1 A metal M and its compound can give the following observable changes in a consequence of reactions



Q.2 **Correct** option regarding the following chemical reaction : (State T for True and F for False)

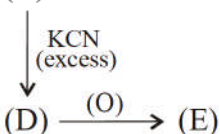
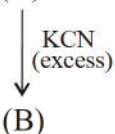


- (i) Complex 'X' is sp^3 hybridised
(ii) ppt. 'Y' is yellow in colour
(iii) Complex 'X' is an type of imperfect complex
(iv) Complex 'X' is an diamagnetic complex

(A) TTTT (B) TTTF (C) FTTF (D) FFTT

Q.3 Concentrated nitric acid reacts with iodine to give

(A) HOI (B) HI (C) HOIO₂ (D) HOIO₃



Colour of ABCDE are respectively.

- (A) Green, Yellow, Reddish Brown, Brown, Yellow
(B) Yellow, Green, Reddish Brown, Yellow, Brown
(C) Brown, Green, Reddish Brown, Brown, Yellow
(D) Green, Reddish Brown, Brown, Yellow, Yellow

Q.5 White compound (A) $\xrightarrow{\text{H}_2\text{S}/\text{dil.HCl}}$ Black ppt. (B) $\xrightarrow{\text{Na}_2\text{S}}$ Black ppt.(C).

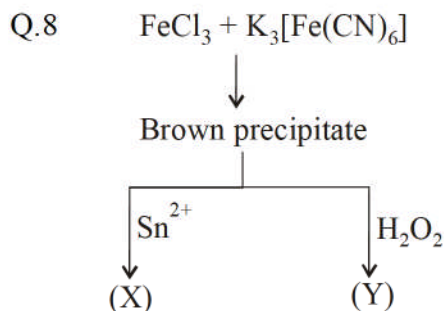
The compound (B) is

(A) Hg (B) Hg₂Cl₂ (C) HgS (D) Hg + HgS

Q.6 From the given mixture of Cr^{3+} , Fe^{3+} and Sn^{4+} . How to separate Fe^{3+} from the other two cations in one step.

(A) dil. HCl + H₂S (B) excess NaOH (C) (NH₄)₂S (D) Excess NH₄OH

- Q.7 When $(\text{NH}_4)_2\text{S}$ is added into neutral FeCl_3 solution, the ppt. is due to formation of
(A) FeS (B) Fe_2S_3 (C) $\text{FeS} + \text{S}$ (D) $\text{Fe}_2\text{S}_3 + \text{S}$



Choose the correct statements

- (A) Both (X) and (Y) are same products, having blue colour
(B) Both (X) and (Y) are different products having same colour
(C) Both (X) and (Y) are different products having different colour
(D) H_2O_2 and Sn^{2+} both act as oxidising agents in given sequence of reactions.
- Q.9 Unknown sample (A) on heating swells up first and aqueous solution of (A) gives white ppt with NaOH which becomes soluble with excess of NaOH addition.
(A) Borax (B) Alum (C) Microcosmic salt (D) None

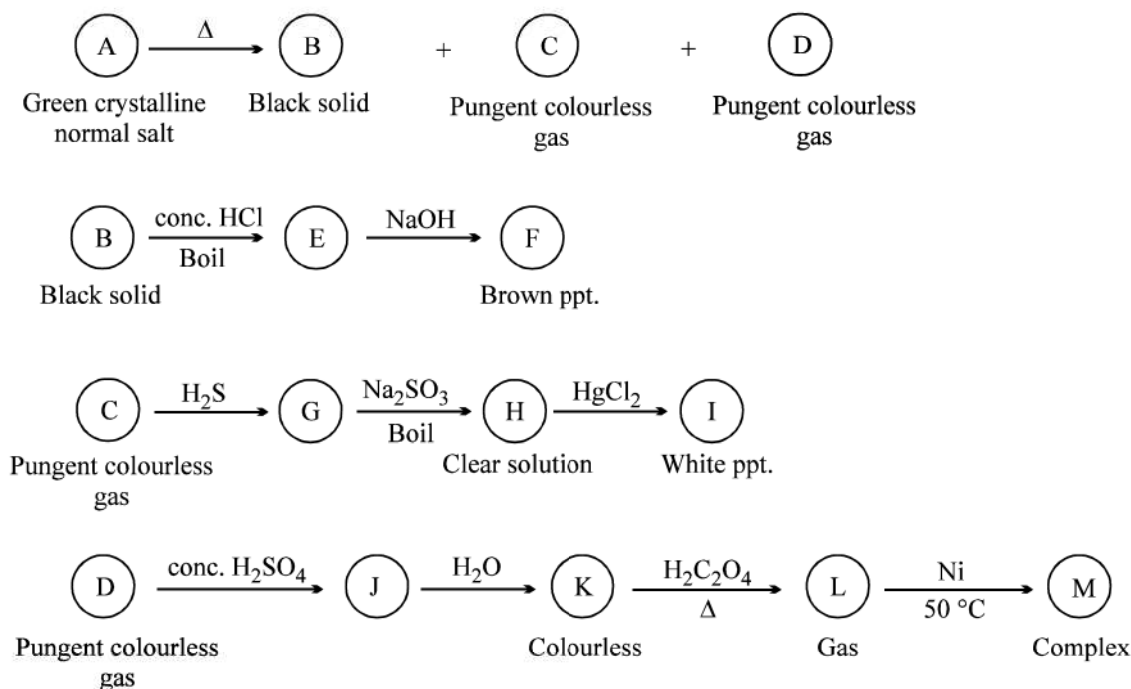
Comprehension

Question No. 10 to 12 (3 questions)

A white crystalline solid 'A' on boiling with caustic soda solution gives a gas 'B' which on passing through an alkaline solution of potassium tetraiodomercurate(II) solution gives a brown ppt. The substance 'A' on heating evolves a neutral gas 'C' which rekindled a glowing splinter but does not give brown fumes with nitric oxide

- Q.10 The gas 'B' is
(A) H_2S (B) NH_3 (C) HCl (D) CO_2
- Q.11 The gas 'C' is
(A) N_2O (B) O_2 (C) NO (D) O_3
- Q.12 The substance 'A' is
(A) NH_4Cl (B) NH_4NO_3 (C) NH_4NO_2 (D) NaNO_3

Question No. 13 to 15 (3 questions)



- Q.13 Type of hybridization of complex (M) is
 (A) sp^3d^2 (B) d^2sp^3 (C) dsp^2 (D) sp^3
- Q.14 Water of crystallization in salt (A)
 (A) 5 (B) 6 (C) 7 (D) 10
- Q.15 Compound (H) + X + $\text{FeCl}_3 \longrightarrow$ Blood Red Colouration.
 The unknown compound X is?
 (A) NaOCN (B) NaCN (C) NaNC (D) $\text{Na}_2\text{S}_2\text{O}_3$

Question No. 16 to 18 (3 questions)

A white solid A reacts with dilute H_2SO_4 to produce a colourless gas B and a colourless solution C. The reaction between B and acidified dichromate yields a green solution and a slightly coloured precipitate D. The substance D, when burnt in air, gives a gas E which reacts with B to yield D and a colourless liquids. Anhydrous copper sulphate turns blue with this colourless liquid. The addition of aqueous NH_3 or NaOH to C produces a precipitate that dissolves in an excess of the reagent to form a clear solution.

Answer the following questions based on above passage.

- Q.16 Which of the following gases are B and E respectively?
 (A) CO_2 and SO_2 (B) SO_2 and H_2S (C) H_2S and SO_2 (D) CO_2 and H_2S
- Q.17 What would appear if the gas B is passed through an aqueous solution of $\text{Pb}(\text{NO}_3)_2$?
 (A) white precipitate soluble in hot dilute HNO_3
 (B) A black precipitate soluble in hot dilute HNO_3
 (C) A black precipitate insoluble in hot dilute HNO_3
 (D) A yellow precipitate soluble in hot concentrated HNO_3
- Q.18 Suppose the solution obtained by the treatment of the solution C with an excess of NaOH is acidified with acetic acid and the gas B is passed through it. Which of the following will be obtained?
 (A) Colourless solution (B) Yellow precipitate
 (C) Black precipitate (D) White precipitate

Question No. 19 to 21 (3 questions)

Using dil. H_2SO_4 gas evolution takes place from several acid radicals. If we consider their smell, colour and other observation if any, several acid radicals can be detected in this step.

Q.19 Using dil. H_2SO_4 , which of the acid radical can not be confirmed.

- (A) NO_2^- (B) CH_3CO_2^- (C) HSO_3^- (D) S^{2-}

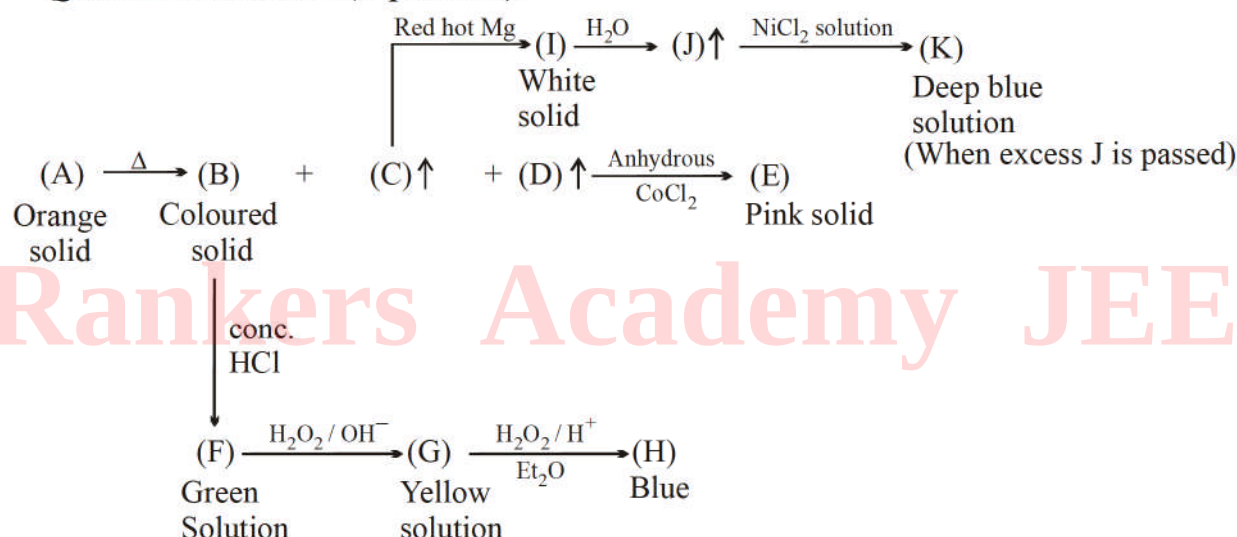
Q.20 When the aqueous suspension of Ag_2SO_3 and Ag_2CO_3 are heated, the respective ppts. are obtained as

- (A) Ag_2O & Ag_2O (B) Ag_2O & Ag (C) Ag & Ag_2O (D) Ag & Ag

Q.21 Which of the following radical(s) is/are producing same gas on treatment with ($\text{Zn} + \text{dil. H}_2\text{SO}_4$).

- I. SO_3^{2-} II. HSO_3^- III. S^{2-} IV. Cl^-
(A) I and II only (B) I, II and III only (C) I, II, III and IV (D) II, III and IV only

Question No. 22 to 24 (3 questions)



Q.22 Which of following gas is turning the filter paper dipped into ($\text{MnSO}_4 + \text{H}_2\text{O}_2$) solution brownish black.

- (A) gas 'C' (B) gas 'D' (C) gas 'J' (D) None of these

Q.23 Type of hybridisation in compound (K)

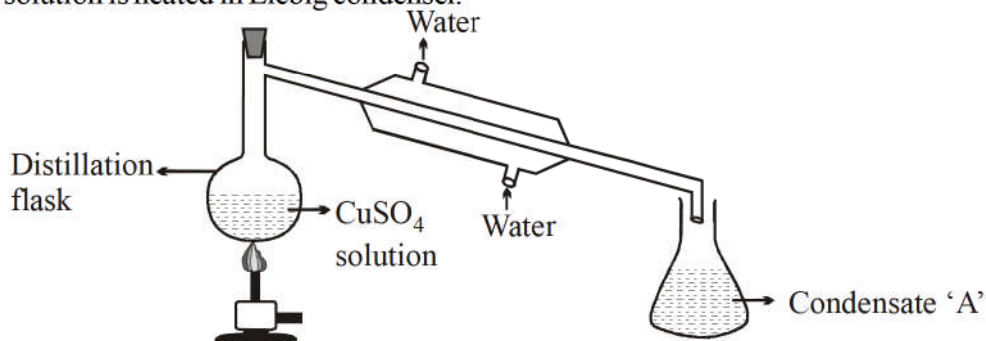
- (A) d^2sp^3 (B) sp^3d^2 (C) dsp^2 (D) sp^3

Q.24 Which of the following set of reagent is suitable to get (F) from (G).

- (A) Neutral H_2O_2 solution (B) $\text{H}_2\text{O}_2 / \text{H}^+$ + amyl alcohol
(C) $\text{H}_2\text{O}_2 / \text{OH}^-$ (D) $\text{H}_2\text{O}_2 / \text{H}^+$

Question No. 25 to 27 (3 questions)

CuSO_4 solution is heated in Liebig condenser.



When condensate 'A' is obtained, white powder 'B' is also obtained in the distillation flask.

- Q.25 The colour of condensate 'A' is
 (A) Blue (B) Yellow (C) Colourless (D) Green
- Q.26 'A' is added to KI solution, which of the following observation is correct.
 (A) Dark brown solution is obtained
 (B) Brownish yellow solution is obtained
 (C) It s being diluted only without any colour change
 (D) None of these is correct.
- Q.27 When white powder 'B' is added to excess of KI solution. Which of the following phenomenon is associated with it.
 (A) Initially redox reaction takes place.
 (B) Dark brown appearance is observed.
 (C) White precipitate is formed which remains invisible by the brown colouration.
 (D) All are correct.

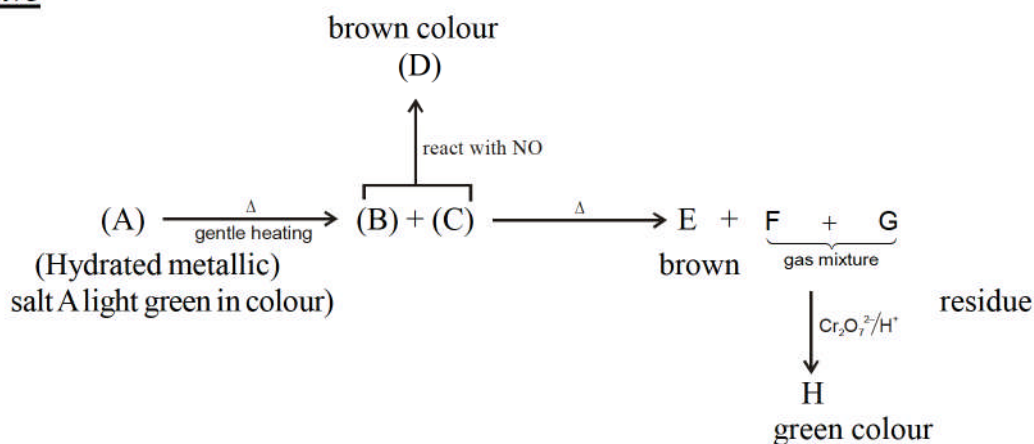
More than one may be correct

- Q.28 Which of the following radical(s) on reaction with dil HCl liberate(s) gas which decolourize(s) acidified KMnO_4 solution.
 (A) $\text{S}_2\text{O}_3^{2-}$ (B) NO_2^- (C) Br^- (D) HCO_3^-
- Q.29 Which of the following reaction is an example of comproportionation reaction.
 (A) $5\text{I}^- + \text{IO}_3^- \xrightarrow{6\text{H}^+} 3\text{I}_2 + 3\text{H}_2\text{O}$ (B) $\text{SO}_2 + 2\text{H}_2\text{S} \rightarrow 3\text{S} + 2\text{H}_2\text{O}$
 (C) $\text{Cl}_2 + \text{OH}^- \rightarrow \text{Cl}^- + \text{OCl}^- + \text{H}_2\text{O}$ (D) $\text{Cl}^- + \text{OCl}^- + 2\text{H}^+ \rightarrow \text{Cl}_2 + \text{H}_2\text{O}$
- Q.30 Which of the following sulphides would dissolve in hot and concentrated nitric acid?
 (A) ZnS (B) HgS (C) MnS (D) Bi_2S_3
- Q.31 Cu^{2+} and Cd^{2+} can be distinguished by which of the following reagent(s).
 (A) Excess KCN solution followed by H_2S reagent
 (B) KI (excess)
 (C) H_2S in dil. HCl medium
 (D) $\text{K}_4[\text{Fe}(\text{CN})_6]$
- Q.32 Which of the following metal produces N_2O gas with 20% HNO_3 .
 (A) Fe (B) Cu (C) Zn (D) Sn
- Q.33 Which of the following acid radicals form white ppt. with AgNO_3 solution.
 (A) $\text{S}_2\text{O}_3^{2-}$ (B) CH_3CO_2^- (C) $\text{C}_2\text{O}_4^{2-}$ (D) HPO_4^{2-}
- Q.34 Which of the following can be used to distinguish between CO_3^{2-} and HCO_3^- ?
 (A) MgCl_2 (B) HgCl_2 (C) $\text{Pb}(\text{CH}_3\text{COO})_2$ (D) AgNO_3

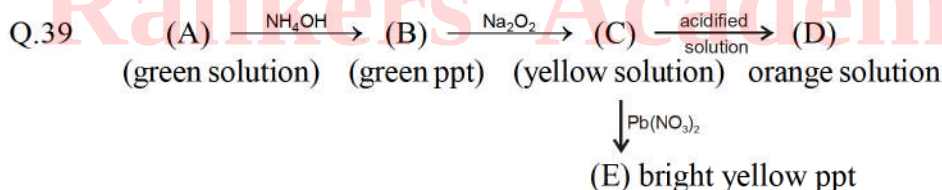
- Q.35 Which of the following reaction are example of redox reactions?
 (A) $\text{KMnO}_4 + \text{H}_2\text{SO}_4 + \text{CO} \longrightarrow$ (B) $\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 + \text{HCl} \longrightarrow$
 (C) $\text{CuSO}_4 + \text{KCN}(\text{excess}) \longrightarrow$ (D) $\text{CuSO}_4 + \text{KI}(\text{excess}) \longrightarrow$
- Q.36 $\text{FeCl}_3 + \text{K}_3[\text{Fe}(\text{CN})_6] \longrightarrow (\text{X}) \xrightarrow{\text{R}}$ Prussian blue colouration.
 'R' could be
 (A) SnCl_2 (B) $\text{H}_2\text{O}_2 + \text{KOH}$ (C) acidified KMnO_4 (D) concentrated H_2SO_4
- Q.37 Select correct options regarding given salt reactions:
 (A) CO_3^{2-} salt + $(\text{CH}_3\text{COO})_2\text{Pb}$ solution $\longrightarrow$ White precipitate
 (B) Cl^- salt + dilute $\text{H}_2\text{SO}_4 \longrightarrow$ White fumes
 (C) S^{2-} salt + Moist $(\text{CH}_3\text{COO})_2\text{Pb} \longrightarrow$ Black precipitate
 (D) SO_3^{2-} salt + $\text{KMnO}_4 / \text{H}^+ \longrightarrow$ Purple colouration

Subjective

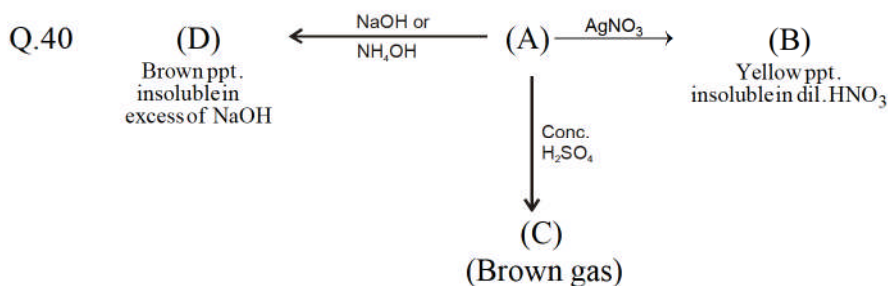
Q.38



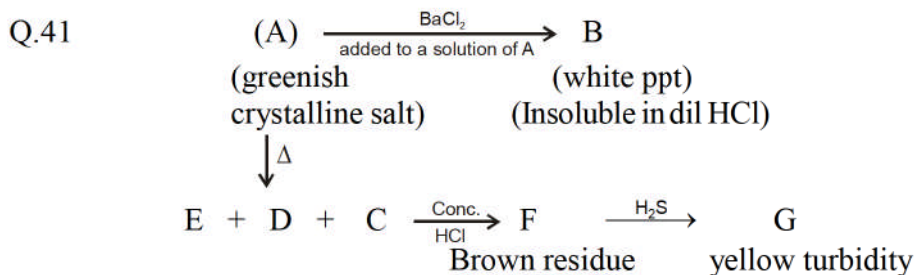
Identify (A) to (H).



Identify (A) to (D).



Identify (A) to (D).



Identify (A) to (G).

ANSWER KEY

Q.1	C	Q.2	A	Q.3	C	Q.4	A	Q.5	D
Q.6	B	Q.7	C	Q.8	A	Q.9	B	Q.10	B
Q.11	A	Q.12	B	Q.13	D	Q.14	C	Q.15	B
Q.16	C	Q.17	B	Q.18	D	Q.19	C	Q.20	C
Q.21	B	Q.22	C	Q.23	B	Q.24	D	Q.25	C
Q.26	C	Q.27	D	Q.28	A	Q.29	ABD	Q.30	ACD
Q.31	ABCD	Q.32	AC	Q.33	ABC	Q.34	ABCD		
Q.35	ABCD	Q.36	AB	Q.37	AC	Q.38	A,B,C,D,E,F,G,H		
Q.39	ABCD	Q.40	ABCD	Q.41	A,B,C,D,E,F,G				

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